Block practice · Radians, sectors, triangles and 3D trigonometry

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Regions made by tangents and a sector

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Practice workbook · Regions made by tangents and a sector

Regions formed by tangents to a circle and a sector, using the right angle between radius and tangent to find lengths and angles.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-06

1. Question 4 · part whole · 6 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part whole only.

4

Diagram **NOT** accurately drawn

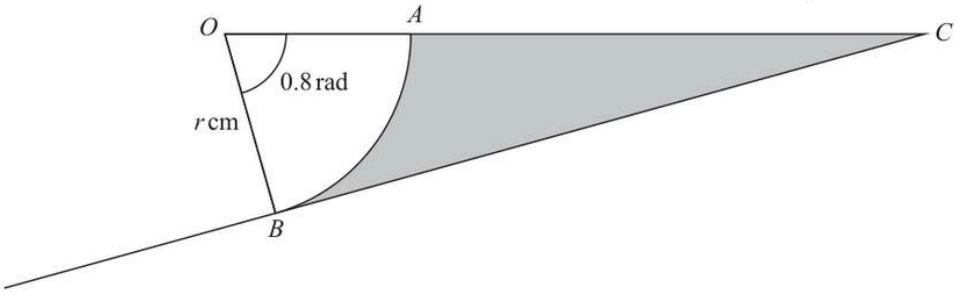


Figure 1

Figure 1 shows a sector AOB of a circle with centre O and radius r cm and a triangle BOC.
The angle of sector AOB is 0.8 radians.
The points O, A and C lie on a straight line so that CB is the tangent to the circle at B.

Given that the area of the shaded region in Figure 1 is 101 cm^2, find the value of r.
Give your answer correct to 3 significant figures.

(6)

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P 5 9 9 3 8 A 0 1 0 3 6

1. Question 4 · part whole · 6 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 6 marks)

1. Question 4 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 5 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

5

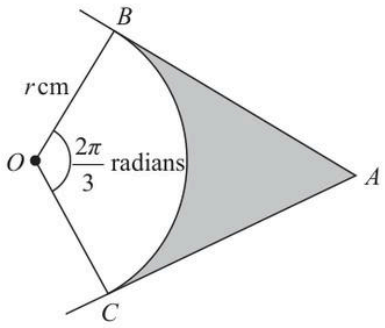


Diagram **NOT** accurately drawn

Figure 2

In Figure 2, AB and AC are tangents to a circle with centre O and radius r cm.

The points B and C lie on the circle so that OBC is a sector of this circle and $\angle BOC = \frac{2\pi}{3}$ radians.

Given that the area of the shaded region is $10\,\text{cm}^2$,

find, to 3 significant figures, the value of r. **(8)**

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2. Question 5 · part whole · 8 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 5 continued

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(Total for Question 5 is 8 marks)

2. Question 5 · part whole · 8 marks · Working and answer

Working and answer

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3. Question 3 · part b.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

3

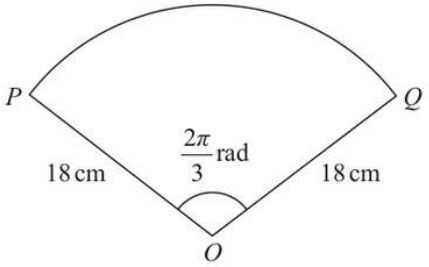


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sector OPQ of a circle with centre O.

The radius of the circle is 18 cm and the angle POQ is $\frac{2\pi}{3}$ radians.

(a) Find the length of the arc PQ, giving your answer as a multiple of π

(2)

Figure 2 below shows the sector OPQ and the kite $OPTQ$.

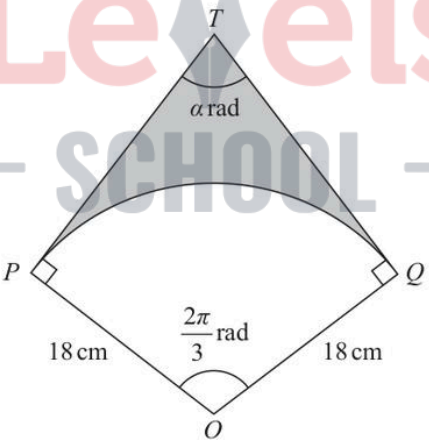


Diagram **NOT** accurately drawn

Figure 2

PT is the tangent to the circle at P and QT is the tangent at Q, such that angle $PTQ = \alpha$ radians.

(b) (i) Find α in terms of π

(1)

(ii) Calculate, to 3 significant figures, the area of the region, shown shaded in Figure 2, which is bounded by the arc PQ and the tangents PT and QT.

(6)

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3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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4. Question 3 · part b.ii · 6 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

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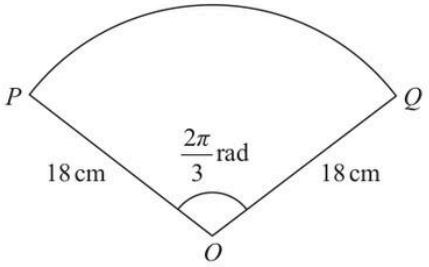


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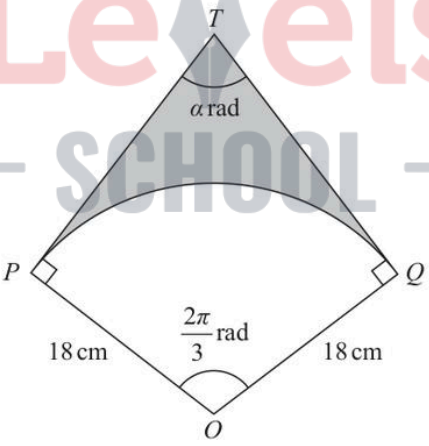


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4. Question 3 · part b.ii · 6 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

Question 3 continued

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4. Question 3 · part b.ii · 6 marks

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Further practice · Attempt part b.ii only.

Question 3 continued

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P 6 6 0 2 5 A 0 8 3 6

4. Question 3 · part b.ii · 6 marks

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Further practice · Attempt part b.ii only.

Question 3 continued

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(Total for Question 3 is 9 marks)

4. Question 3 · part b.ii · 6 marks · Working and answer

Working and answer

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5. Question 3 · part a · 2 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

3

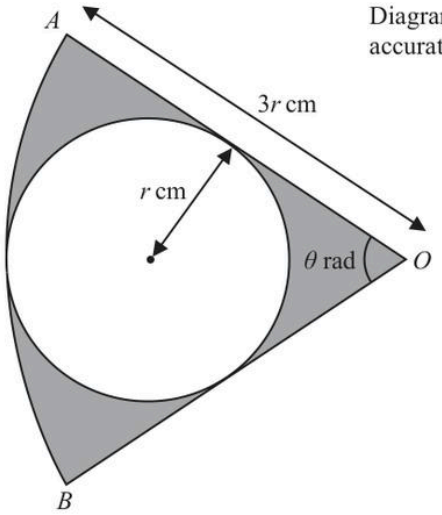


Figure 1

Figure 1 shows the sector AOB of a circle with centre O and radius $3r$ cm

A circle with radius r cm touches OA and OB and the arc AB

Angle AOB is θ radians, where $0 < \theta < \frac{\pi}{2}$

(a) Find the exact value of θ

(2)

The area of the region shown shaded in Figure 1 is 8π cm^2

(b) Find the value of r

(4)

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5. Question 3 · part a · 2 marks · Working and answer

Working and answer

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6. Question 3 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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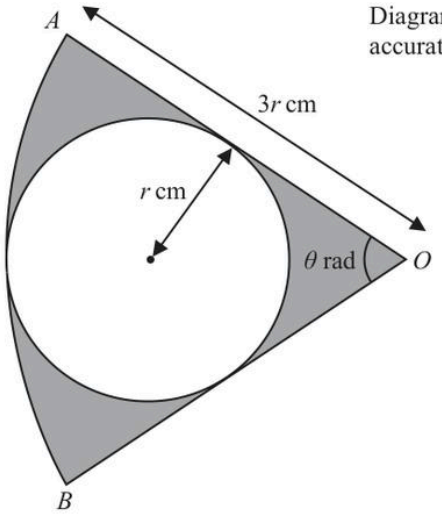


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6. Question 3 · part b · 4 marks

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Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

6. Question 3 · part b · 4 marks · Working and answer

Working and answer

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